EXPERIMENT-04

Aim: Implementing Services in Android. Develop a service that runs in the background and performs a simple task. Observer the life cycle of the service and its interaction with the main application.

Android Timer Service Example
Objective: Develop a Service that runs in the background and performs a timer task. The service will count seconds and show them using Toast.

Theory:
An Android Service is a component that runs in the background without a user interface. It is used for long-running operations such as timers, music playback, or data synchronization. Services continue running even when the user is not interacting with the app.

1. Create Service Class
    - Open Android Studio
    - Right-click package → New → Service → TimerService
    - Implement:
        - Use a thread to run a timer in background
        - Increment seconds every 1 second using Thread.sleep(1000)
        - Display timer using Toast
        - Override lifecycle methods:
    - onCreate() → shows "Service Created"
    - onStartCommand() → starts timer
    - onDestroy() → stops timer
2. Register Service
    - Add in AndroidManifest.xml:

```
<service android:name=".TimerService"/>
```

3. Design Layout
    - Create activity_main.xml with:
        - Start Service button
        - Stop Service button
4. Implement MainActivity
    - Initialize buttons using findViewById()

- Add click listeners:
  - Start button → startService(Intent)
  - Stop button → stopService(Intent)

5. Run Application
   - Click Run
   - Start service to begin timer
   - Stop service to end timer

Code:

TimerService.java

```
activity_main.xml   MainActivity.java   AndroidManifest.xml   TimerService.java

package com.example.experiment_04;

import android.app.Service;
import android.content.Intent;
import android.os.IBinder;
import android.widget.Toast;
public class TimerService extends Service {
    2 usages
    int seconds = 0;
    2 usages
    boolean running = true;
    @Override
    public void onCreate() {
        super.onCreate();
        Toast.makeText( context: this,  text: "Service Created", Toast.LENGTH_SHORT).show();
    }
    @Override
    public int onStartCommand(Intent intent, int flags, int startId) {
        new Thread(new Runnable() {
            @Override
            public void run() {
                while(running)
                {
                    try {
                        Thread.sleep( millis: 1000);
                        seconds++;
                        Toast.makeText(getApplicationContext(),
                                text: "Timer: " + seconds + " seconds",
                                Toast.LENGTH_SHORT).show();
                    } catch (Exception e) {
                        e.printStackTrace();
                    }
                }
            }
        }).start();
        return START_STICKY;
    }
    @Override
    public void onDestroy() {
        running = false;
        Toast.makeText( context: this, text: "Service Stopped",Toast.LENGTH_SHORT).show();
        super.onDestroy();
    }
    @Override
    public IBinder onBind(Intent intent) {
        return null;
    }
}
```

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center"
        android:orientation="vertical">
        <Button
            android:id="@+id/startService"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Start Timer Service"/>
        <Button
            android:id="@+id/stopService"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Stop Timer Service"/>
    </LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```java
package com.example.experiment_04;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.content.Intent;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {
    2 usages
    Button startService, stopService;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        startService = findViewById(R.id.startService);
        stopService = findViewById(R.id.stopService);
        startService.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                startService(new Intent( packageContext: MainActivity.this, TimerService.class));
            }
        });
        stopService.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                stopService(new Intent( packageContext: MainActivity.this, TimerService.class));
            }
        });
    }
}
```

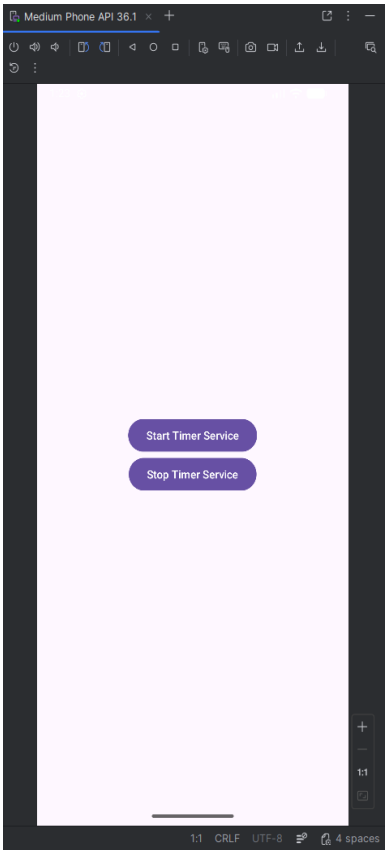
AndroidManifest.xml

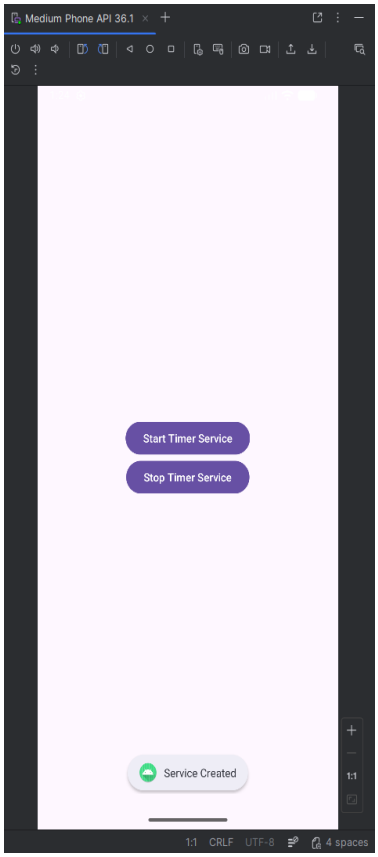
activity_main.xml | MainActivity.java | AndroidManifest.xml | TimerService.java

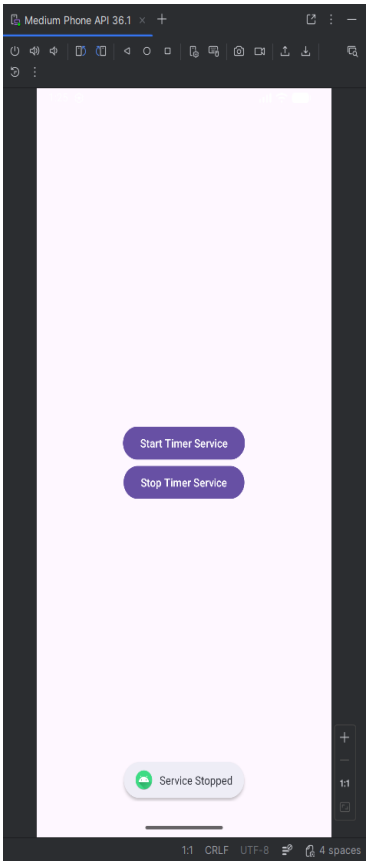
```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="Experiment-04"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Experiment04">
        <service android:name=".TimerService"/>
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Output:

1. Background Execution
   - Service runs independently of UI in the background
2. Timer Functionality
   - Timer increases every second continuously
3. Real-time Output
   - Displays Toast message: Timer: X seconds
4. User Control
   - Start button starts the service
   - Stop button stops the service
5. Service Lifecycle
   - onCreate() → Service initialized
   - onStartCommand() → Timer runs
   - onDestroy() → Service stops







Conclusion

This practical demonstrates how Android Services perform background tasks and how they can be controlled using user interface components.